

4. (1.5 points)

Given the following data and growth curve as of December 31, 2016:

Accident <u>Year</u>	Onlevel Premium <u>(\$000)</u>	Reported Losses <u>(\$000)</u>
2012	1,000	400
2013	1,300	450
2014	1,600	400
2015	1,900	250
2016	2,200	50

- $G(x) = \frac{x^{1.8}}{x^{1.8} + 50^{1.8}}$, where G is the cumulative proportion of ultimate losses reported and x is the average age in months.

Test for expected loss ratio constancy across accident years.

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QUESTION 4

TOTAL POINT VALUE: 1.5

LEARNING OBJECTIVE(S): A2, A3

SAMPLE ANSWERS

Sample 1

AY	Av Age	G(x)	Used Up Premium	AY LR
			=G(x) * OLP	=Rpt Loss / Used up Prem
2012	54	0.5346	534.6	0.7482
2013	42	0.4222	548.86	0.8199
2014	30	0.2851	456.16	0.8769
2015	18	0.1372	260.68	0.9590
2016	6	0.0215	47.3	1.0571

The loss ratio is increasing as AYs get less mature. They are not constant; using a constant ELR does not seem appropriate.

Sample 2

AY	Age	Avg Age	G(x)	LDF	Ult (000)	LR
2012	60	54	0.53	1.887	754.8	0.755
2013	48	42	0.42	2.381	1,071.45	0.824
2014	36	30	0.29	3.448	1,379.2	0.862
2015	24	18	0.14	7.143	1,785.75	0.940
2016	12	6	0.02	50	2,500	1.136

AY 13 Sample Calcs

Avg Age = 48 – 6 = 42

$G(x) = 42^{1.8} / (42^{1.8} + 50^{1.8}) = 0.42$

$LDF = 1 / 0.42 = 2.381$

$Ult = 450 \times 2.381 = 1,071.45$

$LR = 1,071.45 / 1300 = 0.824$

Conclude that ultimate loss ratios are increasing over time (not constant across AYs).

Sample 3

AY	On-level Prem	Reported Loss	Avg Age	G(x)	Rep loss G(x)*EP
12	1M	400k	54	0.535	0.748
13	1.3M	450k	42	0.422	0.820
14	1.6M	400k	30	0.285	0.877
15	1.9M	250k	18	0.137	0.959
16	2.2M	50k	6	0.022	1.056

Total ELR = sum (Reported Loss) / sum (Prem * G(x)) = 1550/1847.5 = 83.9%

Loss ratio by year is not constant – ranges from 74.8% to 105.6%. The overall ELR is 83.9% which will not provide appropriate IBNR for those high/low years. Overestimate in low LR years and

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underestimate in high LR years.

Sample 4

I will assume the LDF method is used

AY	X	G(x)	LDF = $G(x_t)/G(x)$	Ult LR
2012	$60 - 6 = 54$	$54^{1.8} / (54^{1.8} + 50^{1.8}) = 0.535$	$0.815/0.535 = 1.523$	$400 \times 1.523 / 1000 = 60.9\%$
2013	42	0.422	1.931	66.8%
2014	30	0.285	2.860	71.5%
2015	18	0.137	5.949	78.3%
2016	6	0.022	37.045	84.2%

Testing twice the last age for truncation: $G(x_t) = G(2 \times 60 - 6) = G(114) = 0.815$. This is significantly lower than 1.0 so the curve should be truncated at $G(x_t) = G(114)$.

The ultimate expected LRs are increasing with the AY, so there is not expected LR constancy.

Sample 5

AY	X	G(x)	Used-up Prem	Expected Losses	Diff b/w Actual & Expected Losses
	<i>avg age</i>		<i>G(x)*Prem</i>	<i>0.839*G(x)*Prem</i>	
2012	54	0.535	535	449	$400 - 449 = -49$
2013	42	0.422	549	460	-10
2014	30	0.285	456	383	17
2015	18	0.137	260	218	32
2016	6	0.022	48	41	9

A constant ELR likely not appropriate, as actual LR tended to increase over AYs, and is not constant.

Sample 6

X	G(x)
54	0.535
42	0.422
30	0.285
18	0.137
6	0.022

$$ELR = \frac{400 + 450 + 400 + 250 + 50}{1000(0.535) + 1300(0.422) + \dots + 2200(0.022)} = 0.839$$

$$\text{Expected Loss} = (\text{Prem})(ELR)(G(x))$$

$$\text{Resid} = (A - E) / E^{0.5}$$

AY	Exp Loss	Act Loss	Resid
2012	$1000(0.839)(0.535) =$	400	-2.31

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	449		
2013	$1300(0.839)(0.422) = 460$	450	-0.48
2014	$1600(0.839)(0.285) = 382$	400	0.89
2015	$1900(0.839)(0.137) = 218$	250	2.14
2016	$2200(0.839)(0.022) = 41$	50	1.47

Since the resid appear to increase with time, we do NOT observe LR constancy across AY.

EXAMINER'S REPORT

Candidates were expected to know how to calculate $G(x)$ and translate that into either expected ultimate losses or used up premium, and then compare AYs using either a loss ratio or residual approach to verify the constancy of the expected loss ratio.

The most common error was using a Cape Cod approach. Candidates were expected to realize that they were being asked to verify if the underlying expected loss ratio was appropriate, and therefore should not use that expected loss ratio in their calculation of ultimate loss ratio by accident year.

Additionally, some candidates spent time calculating whether the $G(x)$ values should be truncated. Because the question was asking to look at the trend of the ultimate loss ratios, a candidate who truncated the $G(x)$ values spent time adjusting all $G(x)$ values (and therefore the ultimate loss ratios) by the same value, which had no impact on the final observation about the trend of the ultimate loss ratios.